Impulse Response Part II

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1. The impulse response of two systems connected in parallel is the _____ of the individual system impulse responses.

a) sum

b) convolution

c) integral

d) derivative

2. The impulse response of two systems connected in series, i.e., cascade, is the _____ of the individual system impulse responses.

a) sum

b) convolution

c) integral

d) derivative

3. If the absolute value of all the values in the impulse response sum to a finite number, the system guaranteed to be _____ .

a) causal

b) stable

c) memoryless

d) finite impulse response

4. Three LTI systems have impulse responses $h_1[n], h_2[n]$, and $h_3[n]$. They are connected to obtain an LTI system with impulse response $h[n] = h_1[n] * (h_2[n] + h_3[n])$. Which of the following interconnections are equivalent to $h[n]$?

a) $h_2[n] * (h_1[n] + h_3[n])$

b) $h_2[n] * h_1[n] + h_1[n] * h_3[n]$

c) $h_1[n] * h_2[n] * h_3[n]$

d) $(h_2[n] + h_3[n]) * h_1[n]$

e) $h_2[n] * h_1[n] + h_3[n] * h_1[n]$

f) $h_2[n] * h_1[n] + h_2[n] * h_3[n]$

5. A continuous-time system has impulse response $h(t) = \cos(10\pi t)u(t)$. Is this system stable?

6. A continuous-time system has impulse response $h(t) = u(t) - u(t-5)$. Is this system stable?

Download `Commute.m` and `LTI.mat` from the website. `Commute.m` processes a short audio clip in two ways using systems with two different impulse responses, `h` and `g`. First it convolves `song` with `h` followed by `g`. Next, it convolves `song` with `g` followed by `h`. If convolution is commutative, these two procedures should yield identical audio clips. `Commute.m` plots the first 50 values of `h`, `g`, `h*g` and the difference between `song*h* g` and `song*g*h`. It also plays the original clip `song`, `song*h`, `song*g`, `song*h*g`, and `song*g*h`. If convolution is commutative, then the difference between `song*h*g` and `song*g*h` should be zero (or numerically very small) and `song*h*g` and `song*g*h` should sound the same.

7. What signal is the best description for $h[n] * g[n]$

a) $u[n]$

b) $(0.9)^n u[n]$

c) $\sin(\pi n) u[n]$

d) $\delta[n]$

e) $\cos(\pi n) u[n]$

8. Based on the answer to the previous question, the cascade of $h[n]$ and $g[n]$ _____ .

a) attenuates low frequencies

b) enhances low frequencies

c) leaves the signal unchanged

d) attenuates high frequencies

e) enhances high frequencies